01 / COVER + PROBLEM STATEMENT

Solar hai. Ab smart use kaise ho?

Green Energy Management | Renewable energy ko monitor aur optimize karne ka practical proposal.

UrjaSetu

Existing solar ko replace nahi: supported data, useful action aur extra bachat ka transparent proof.

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Problem statement, simple words mein

Aisa solution banana hai jo renewable energy ki **production aur consumption track** kare, useful scheduling suggest kare, carbon impact estimate kare aur supported grid signals se integrate ho. Dashboard tab useful hai jab user ko samajh aaye: **ab kya karna hai aur kyun?**

PS requirement	UrjaSetu ka response
Consumption tracking	W / kW = abhi ka power; kWh = time ke saath used energy. Freshness aur source label ke saath.
Renewable optimization	Flexible task ko suitable solar / tariff slot mein schedule; deadline aur essential loads protect.
Carbon footprint	Grid-use-based operational CO2 estimate; factor, year aur assumptions visible. [6,7]
Smart grid integration	Demo: simulated tariff / grid events. Real utility access aur approved control future pilot mein.

Kis user se start karna chahiye?

Initial target: existing-solar homes, hostels ya small businesses with flexible loads. Retrofit usefulness aur affordability abhi hypotheses hain; interviews aur pilots se verify karna hoga.

STATUS: proposal, not a proven commercial product

Market mein AI energy products already hain [4,5]. Hamare savings, compatibility aur reliability ko build aur pilot testing se prove karna hoga.

02 / PROPOSED SOLUTION

Existing setup, smarter management

Solar panel, inverter aur existing electrical protection retain. Pehle compatibility aur benefit check.

Do practical connection routes

A. Existing data available

Owner-approved API / Modbus read-only adapter. Model aur available fields verify. Fronius Solar API ek real example hai [1]. Missing data ke liye hi extra meter.

B. Data missing / closed system

Suitable external AC energy meter use karo. CT + voltage measurement chahiye; sirf current se accurate kWh nahi. Internal faults / battery SOC milna guaranteed nahi. [2]

Hardware kahan lagega?

Hardware	Installation location + purpose
Gateway box	Dry enclosure near DB / inverter; isolated low-voltage supply. Demo ESP32; real gateway tested hardware. RTU ho to isolated RS485 interface.
Energy meter + CTs Only if needed	Simple single-phase, no-battery site: grid connection par import/export; inverter AC feeder par PV output. Suitable 2-channel meter + 2 CTs is one option. [2]
Optional load actuator	Selected appliance / DB par load-rated smart plug, ya motor-rated contactor + protection. Gateway command deta hai; home power carry nahi karta.
Network + measurement	Wi-Fi / LAN via router; load-wise exact use ke liye sub-meter / metered plug. 3-phase aur hybrid sites ko separate metering design chahiye.

Installation se useful action tak

1. Site survey + owner permission. **2.** Qualified electrician all sources isolate karke approved metering / actuator install kare. **3.** Gateway pair karo, import/export direction aur units verify karo. **4.** Pehle monitoring baseline; phir useful plan par user opt-in.

Important limits

Grid-tied solar self-consumption pehle se hota hai; ham flexible loads ka time improve karte hain. Hybrid AC output ko pure solar na bolo. No meter-seal tampering, DIY mains switching or bypass of inverter protection.

03 / FLOW OF SOLUTION

Data se decision, phir verified action

Blue: readings | Purple: planning | Teal: approved control | Red: hold / safe fallback

1. CONNECT + READ

Supported inverter / meter -> gateway -> secure backend. Device ID, timestamp, W / kWh aur status store karo.

2. Readings fresh aur valid hain?

Units, range, missing values aur connection check. Measured, estimated aur simulated data separate.

NO

HOLD

Stale / faulty data ko label karo. Naye auto commands block; connection recover karke retry.

YES

3. FORECAST + TASK PLAN

History / optional weather se solar-demand estimate. Tariff + export credit + deadline + runtime ke basis par useful schedule choose karo.

AI ka role

Backend: forecast. Low history: rules. LLM: explanation, not electrical safety.

Useful, feasible aur authorized?

Benefit positive? Task limits valid? User approval / pre-approved rule + supported actuator + local safety checks?

NO

NO ACTION

No actuator: advice only. No gain: monitor only. Conflict: explain aur revise.

YES

4. APPLY APPROVED COMMAND

Backend -> gateway -> rated actuator -> selected load. Ack aur actual power verify; critical loads protected.

5. RE-MEASURE + SAVINGS RECEIPT

Energy log, task result, baseline-based extra saving aur CO_2 estimate app mein show. Fresh data se next plan.

OFFLINE

Fresh local data + valid approved plan: continue. Warna safe fallback; buffer readings. Manual override safety bypass nahi kare.

04 / TECH STACK + IMPLEMENTATION

24 hours: one reliable end-to-end demo

Suggested stack; team familiarity ko priority. Forecasting optional hai, core control loop compulsory.

Layer	Recommended implementation
Firmware / telemetry	ESP32 C++ (ESP-IDF or Arduino); DC sensor over I2C. Send readings + status. INA219 is a DC monitor, not a mains meter. [8]
Transport / backend	MQTT over TLS or HTTPS; FastAPI for ingestion and command validation. Per-device credentials, timestamps, command IDs + expiry. [3]
Storage / frontend	SQLite for demo; PostgreSQL for pilot. React responsive web dashboard; Python-heavy team can use Streamlit for demo.
AI / optimization	Start with transparent rules + enumerate valid time slots. Later forecast with a simple regression model; compare to a naive baseline before using it.
User / security	Hinglish explanation, approve / skip, estimated savings, last update. Role-based access, audit logs; no public unauthenticated control endpoint.

App flow: overview -> suggestion -> approve/skip -> verified status -> Savings Receipt. In-app alerts are core; opt-in background notifications are optional and depend on permission / connectivity.

Minimum low-voltage demo kit + budget

Component / quantity	Budget basis
ESP32 development board x1	Rs 489 listed incl. GST, Robocraze [9].
INA219 DC sensor x1	Rs 98 listed incl. GST, Robu [10]. Module ratings verify.
DC load + suitable driver	Rs 250 planning allowance; motor ho to flyback protection.
Protected 5-12 V supply + cable	Rs 250 planning allowance; match voltage / current.
Fuse, wires, terminals, board, enclosure	Rs 300 planning allowance. Lab multimeter borrow karo.

Planning subtotal: Rs 1,387 (489 + 98 + 250 + 250 + 300). **Target reserve: Rs 1,500 + shipping.** Sirf first two prices observed listings hain; baaki estimates. Laptop, phone, test tools, optional mini-panel / second sensor excluded. Real-home installed price alag quote hoga.

Build order / 24-hour team plan

0-4h: safe bench + sensor check. **4-10h:** telemetry + database + UI.
10-18h: rules, approval, actuator + receipt. **18-24h:** fault tests, repeat demo, evidence video.

05 / USP + USER BENEFIT

Our edge must be useful, not just new

Proposed differentiation: focused retrofit + task completion + transparent incremental benefit.

The value proposition

"Aapka existing solar setup rakho. Supported readings ko jodo, kaam sahi waqt par chalao, aur extra bachat ka calculation dekho." Yeh positioning hai; market uniqueness abhi validated nahi.

Proposed USP	User ko kya milega / proof kya chahiye
Eligibility before purchase	Pehle flexible load, tariff aur data access check. Extra gain useful na ho to hardware recommend nahi. Prove with site assessment.
Task-first scheduling	"Pump 30 min, 4 PM se pehle" - sirf ON/OFF toggle nahi. Deadline, minimum runtime, critical loads aur manual preference preserve.
Supported retrofit + clarity	Tested adapters ka compatibility list; same data on mobile/web. Simple reason, approve/skip, offline/last-update label.
Savings Receipt	Measured energy + same-work baseline + assumptions. Existing solar ki total savings ko optimizer ki savings mat bolo.

Saving calculate kaise hogi?

Operating cost = import charges - export credit + relevant operating costs.
Extra saving = comparable baseline cost - optimized cost. Same task aur comparable solar conditions use karo. Hardware / installation / service cost se payback alag test karo; battery ho to losses count once.

Illustration only - current tariff ya guarantee nahi

1 kWh task grid se solar-surplus slot mein shift: import Rs 8/kWh, export Rs 3/kWh. Gross benefit Rs 5, not Rs 8; other costs subtract. Export compensation same ho to gain zero bhi ho sakta hai.

Carbon report aur mobile explanation

Grid CO_2 estimate = imported kWh x selected CEA factor (kg CO_2/kWh); factor year / boundary show karo [6]. Yeh operational estimate hai, full lifecycle footprint ya certified credit nahi. Timing shift alone se carbon reduction assume mat karo; avoided-emissions claims separately evaluate karo [7].

App message: "Solar available hai; 30-min pump task ab schedule karein?" + reason + approve/skip.

06 / FEASIBILITY + COMPETITORS

Narrow use case. Measurable value.

Feasibility = available time, skills, budget aur constraints ke andar solution bana aur test kar paana.

Alternative	Already available	How we should respond
SolarEdge ONE [4]	AI optimization for using, storing and selling energy; preferences and connected devices.	Never claim AI dashboard is new. Target supported sites where an add-on has proven incremental value.
Schneider Wiser [5]	Monitoring / control and predictive scheduling of supported large loads.	Compete with clear setup, local explanation and task-focused pilot; not unsupported superiority claims.
Smart / energy meters [2]	Power / energy visibility; some models have networking and automation interfaces.	Reuse a meter where appropriate. Meter data access is model-specific; a utility meter API is not guaranteed.
Basic inverter / manual setup	Generation display or basic app; capabilities vary widely.	Eligibility check: add advice / supported control only if existing features do not already solve the task.

Assessment: 24h demo is realistic only with ready hardware + a familiar stack. Real-home readiness needs a separate pilot.

Market caveat: features, prices and availability vary by country / model. No claim that competitors lack alerts, apps or automation. Sources demonstrate feasibility of technology, not UrjaSetu product performance.

Practical feasibility + risk handling

Risk / constraint	Plan and acceptance check
24-hour scope	One real DC load, one sensor, one user flow. Grid / tariff / PV replay labelled. Freeze extras if the core loop is unstable.
Safety / reliability	No mains demo. Test stale sensor, offline network, denied approval, command failure and manual override. No unsafe start; never report failed action as success.
AI / data quality	Rules first; chronological forecast validation later. Report error vs naive forecast; anomalies need human diagnosis, not a guaranteed fault claim.
Real-home adoption	Qualified installer, compatible metering, electrical protection, warranty / approvals and security review. No universal plug-and-play claim.

Go-to-market: validate before scaling

Proposed next step: 10 user interviews + 3 consented pilot sites, about 4 weeks. Compare same-work baselines with recorded solar/tariff conditions; track task completion, data gaps, extra cost benefit and installation effort. Partner with installers; offer monitoring first, automation only when justified.

07 / RESEARCH + REFERENCES

Evidence + submission checklist

Primary sources for technical claims. Retail listings support only observed component prices.

[1] Fronius - Solar API (JSON)
Documented network read access; exact inverter compatibility still must be checked.

[2] Shelly - Pro EM-50 knowledge base
CT-based energy metering, voltage inputs and network interfaces; example, not a universal BOM.

[3] Espressif - ESP-MQTT
Official MQTT client supports secure transport; application must implement authentication and checks.

[4] SolarEdge - SolarEdge ONE
Manufacturer-described AI energy optimization; establishes that this feature category already exists.

[5] Schneider Electric - Wiser Home AI
Official announcement of predictive scheduling / automated management for supported loads.

[6] Central Electricity Authority - CO2 database
Select appropriate factor, reporting year and method; do not hard-code an unexplained number.

[7] GHG Protocol - Avoided emissions
Comparative emissions need transparent boundaries, assumptions and a credible counterfactual.

[8] Texas Instruments - INA219
DC current / voltage / power monitoring; module and shunt limits must be verified for the bench.

[9] Robocraze - ESP32 38-pin board
Observed price: Rs 489 including GST. Recheck seller, stock and delivery before purchase.

[10] Robu - CJMCU-219 sensor module
Observed price: Rs 98 including GST. This price is not a complete tested-hardware guarantee.

What is NOT yet proven?

No completed UrjaSetu field trial, certified mains device, verified annual saving, trained forecast benchmark or universal compatibility is evidenced here. Dashboard pictures are design mockups. Next proof: repeatable demo logs + same-work comparison + failure-test video.

Before submission

Team name / final member list: add confirmed details.
GitHub repository + demo link: add real tested links; none supplied yet.
Template mapping: cover + problem merged on page 1; solution, flow, tech, USP, feasibility/competitors and references retained. Guidance slide omitted.